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What are the different logs under kafka data log dir

Asked 4 years, 5 months ago Modified 2 years, 4 months ago Viewed 4k times



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I am trying to understand the kafka data logs. I can see the logs under the dir set in logs.dir as "Topicname_partitionnumber". However I would like to know what are the different logs captured under it. Below is the screenshot for a sample log.



```
00000000000000000217.index
00000000000000000217
00000000000000000217.timeindex
000000000000000004411.index
000000000000000004411
000000000000000004411.timeindex
000000000000000008534.index
000000000000000008534
000000000000000008534.timeindex
000000000000000012664.index
000000000000000012664
000000000000000012664.timeindex
```

apache-kafka

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edited Dec 25, 2020 at 12:08

asked Dec 12, 2018 at 13:56



Ivan Aracki

4,691 11 57 69



Parth Gandhi

311 8 16

Possible duplicate of [why .index file exists in kafka-log directory?](#) – Marius Soutier Dec 12, 2018 at 14:35

3 Answers

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In Kafka logs, each partition has a log.dir directory. Each partition is split into segments.

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A segment is just a collection of messages. Instead of writing all messages into a single file, Kafka splits them into chunks of segments.



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becomes active. One partition can have one or more segment based on the configuration.



Each segment contains three files - **segment.log, segment.index** and **segment.timeindex**

There are three types of file for each Kafka topic partition:

```
-rw-r--r-- 1 kafka hadoop 10485760 Dec 3 23:57 00000000000000000000.index
-rw-r--r-- 1 kafka hadoop 148814230 Oct 11 06:50 00000000000000000000.log
-rw-r--r-- 1 kafka hadoop 10485756 Dec 3 23:57 00000000000000000000.timeindex
```

The 00000000000000000000 in front of log and index files is the name of the segments. It represents the offset of the first record written in that segment. If there are 2 segments i.e. Segment 1 containing message offset 0,1 and Segment 2 containing message offset 2 and 3.

```
-rw-r--r-- 1 kafka hadoop 10485760 Dec 3 23:57 00000000000000000000.index
-rw-r--r-- 1 kafka hadoop 148814230 Oct 11 06:50 00000000000000000000.log
-rw-r--r-- 1 kafka hadoop 10485756 Dec 3 23:57 00000000000000000000.timeindex
-rw-r--r-- 1 kafka hadoop 10485760 Dec 3 23:57 00000000000000000002.index
-rw-r--r-- 1 kafka hadoop 148814230 Oct 11 06:50 00000000000000000002.log
-rw-r--r-- 1 kafka hadoop 10485756 Dec 3 23:57 00000000000000000002.timeindex
```

.log file stores the offset, the physical position of the message, timestamp along with the message content. While reading the messages from Kafka at a particular offset, it becomes an expensive task to find the offset in a huge log file. That's where .index the file becomes useful. It stores the offsets and physical position of the messages in the log file.

Index File		Log File			
Offset	Position	Offset	Position	Time	Message
0	0	0	0	1533443377944	Hello World
1	79	1	79	1533462689974	Amazon

.timeindex the file is based on the timestamp of messages.

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edited Jun 20, 2020 at 9:12

answered Dec 12, 2018 at 14:36



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Nishu Tayal
19.9k 8 48 101

Thanks. This makes it clear. Also i do not see message timestamp in .log file. Do we need to make any changes in the config file to get it recorded? We need to know the message in time in the partition.

– Parth Gandhi Dec 12, 2018 at 15:31

timestamps are included in the messages for Kafka 0.10+ version. Which kafka version are you using?

– Nishu Tayal Dec 12, 2018 at 21:24

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Here is a sample from one of the .log (data) file. I can see few extra characters/symbols but not the date. I tried opening it with notepad and notepad++. pastebin.com/hStyCW13 – [Parth Gandhi](#) Jan 9, 2019 at 12:04 



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The files without a suffix are the segment files, i.e. the files the data is actually written to, named by earliest contained message offset. The latest of those is the active segment, meaning the one that messages are currently appended to.

.index are corresponding mappings from offset to positions in the segment file. .timeindex are mappings from timestamp to offset.

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answered Dec 12, 2018 at 14:34



[Marius Soutier](#)

11.2k 1 37 48

The files without extensions are like that probably because of a Windows configuration option that hides the extension when it is linked to an application – [Raul Lapeira Herrero](#) Sep 11, 2021 at 14:42

Certainly not running Kafka on Windows :) – [Marius Soutier](#) Sep 14, 2021 at 12:26

Check the screenshot from OP, that is Windows Explorer. Kafka runs in Windows, also he could have copied the files with FTP. – [Raul Lapeira Herrero](#) Sep 14, 2021 at 21:20



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Below is the screenshot for a sample log

You should add your screenshot and sample log, then we could give your expected and specific answer.

before that, only can give you some common knowledge:

eg: in my CentOS, for folder:

```
/root/logs/kafka/kafka.log/storybook_add-0
```

- storybook_add : is the **topic** name
- in code, the real topic name is storybook-add

its contains:

```
[root@xxx storybook_add-0]# ll
```

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```
-rw-r--r-- 1 root root 10485756 Aug 28 16:44 0000000000000000023.timeindex
-rw-r--r-- 1 root root          9 Aug 28 16:44 leader-epoch-checkpoint
```

- 0000000000000000023.log : log file
 - store the real data = kafka message
- 0000000000000000023.index : index file
- 0000000000000000023.timeindex : timeindex file

->

- 0000000000000000023 called segment name
 - why is 23 ?
 - in 0000000000000000023.log , which stored first message's position is 23
 - kafka previously has totally received 23 messages
- what the message data look like?
 - we can see through its content:

```
[root@... ~]# pwd
/root/.ssh
[root@... ~]# cd /root/.ssh
[root@... ~]# ll
total 8
-rw-r--r-- 1 root root 10485760 Aug 28 16:44 0000000000000000023.index
-rw-r--r-- 1 root root          9 Aug 28 16:44 0000000000000000023.log
-rw-r--r-- 1 root root 10485756 Aug 28 16:44 0000000000000000023.timeindex
-rw-r--r-- 1 root root          9 Aug 28 16:44 leader-epoch-checkpoint
[root@... ~]# cat 0000000000000000023.log
{"source": "peanut", "id": "5cbed2bc3c51b8319a2d6c31", "action": "UPDATE", "time": "2019-08-28 16:44:37"}{"source": "peanut", "id": "5cbed2bc3c51b8319a2d6c39", "action": "UPDATE", "time": "2019-08-28 16:44:46"}{"source": "peanut", "id": "5cbed2bc3c51b8319a2d6c3b", "action": "UPDATE", "time": "2019-08-28 16:44:53"}M;wLF{"source": "peanut", "id": "5cbed2bc3c51b8319a2d6c3b", "action": "UPDATE", "time": "2019-08-28 16:44:53"}M;wLF
```

For further basic concept and logic of kafka, recommend to read this article:

[A Practical Introduction to Kafka Storage Internals](#)

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answered Aug 28, 2019 at 10:26



crifan

12.6k

1 68 51